WINDSHIELD WIPER ASSEMBLY INSPECTION

AFFECTED VEHICLES: CERTAIN 2001 MODEL YEAR ESCAPE

OVERVIEW

The affected vehicles in this program may require the replacement of the windshield wiper mounting arm and pivot shaft assembly (wiper module assembly less motor). A two-part inspection must be performed to determine if replacement is required. The inspection includes the following:

- An inspection of the wiper link ball studs.
- An inspection of the wiper link plastic ball sockets.

If the vehicle fails the inspection process, a ***new*** windshield wiper mounting arm and pivot shaft assembly must be installed. This procedure gives details of the inspection process and refers you to the service procedures in the Workshop Manual if replacement is required.

INSPECTION

CAUTION: Whenever the wiper arms are removed and reinstalled, proper torquing of the wiper arm nuts is critical.

1. Verify that the wipers are in the “PARK” position and the key is in the “OFF” position before proceeding with inspection.
2. Remove the windshield wiper arm nuts, then remove the wiper arms.
3. Remove the left side cowl top vent panel.
4. Disconnect the wiper motor electrical connector.
5. Remove the three (3) bolts, then remove the wiper module. See Figure 1.

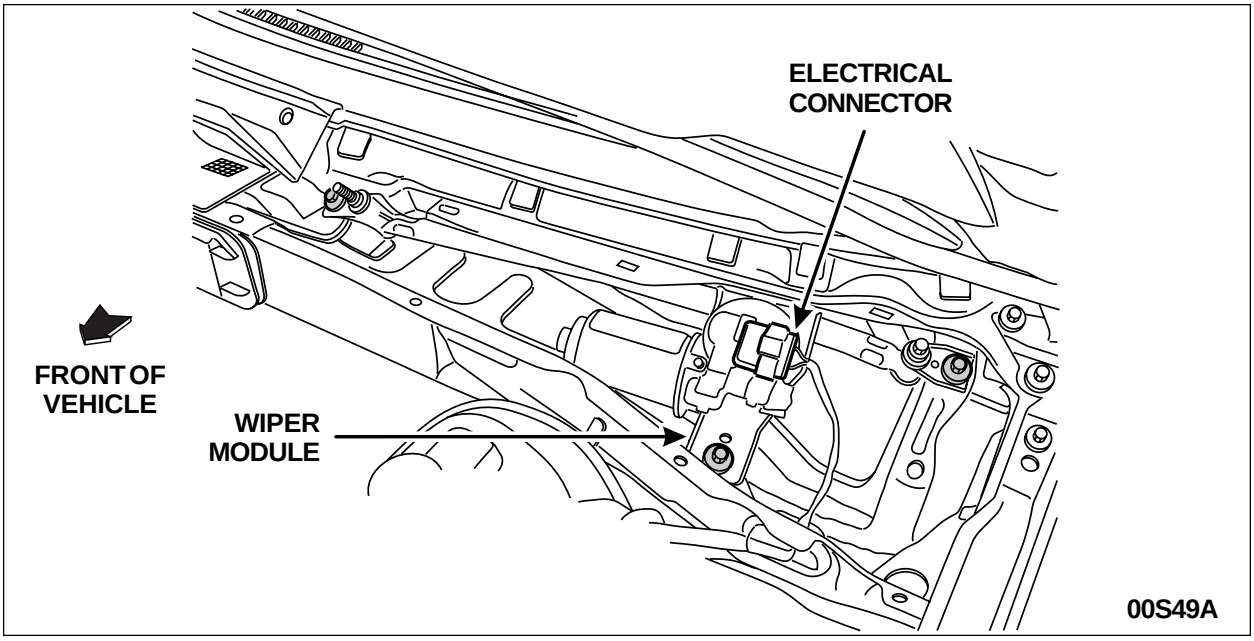
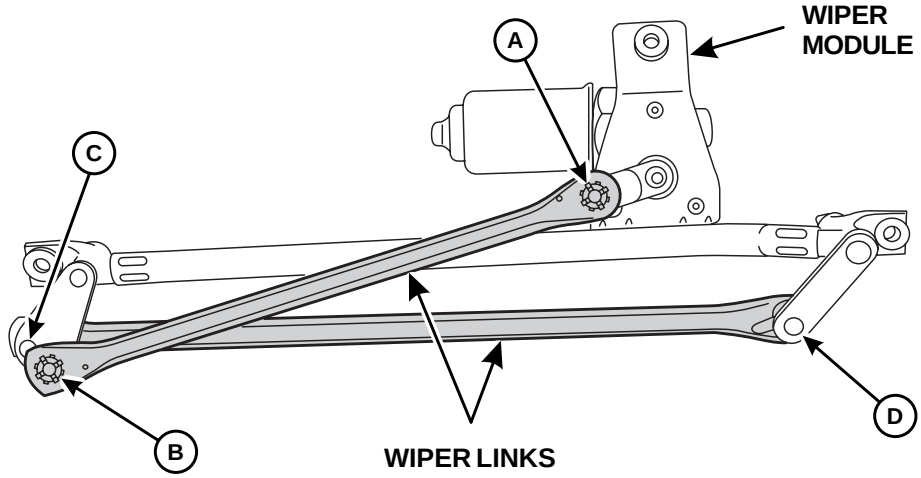


FIGURE 1

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IMPORTANT: USING A PERMANENT MARKER, PLACE A MARK ON THE WIPER LINKS NEXT TO ALL BALL STUDS/SOCKETS THAT FAIL INSPECTION. DEFECTIVE WINDSHIELD WIPER MOUNTING ARM AND PIVOT SHAFT ASSEMBLIES MUST BE RETURNED TO THE WARRANTY PARTS RETURN CENTER. SEE THE ATTACHMENT II FOR MORE INFORMATION.

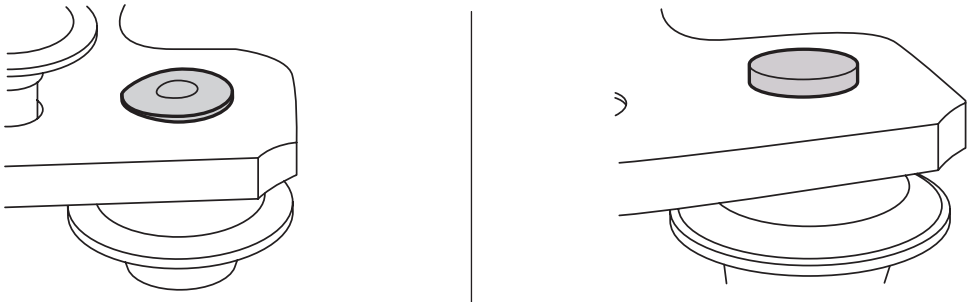
6. Inspect the four (4) ball studs (positions A-D). The backside of each ball stud should be riveted (mushroomed) over as shown in the illustration. Please note, the ball stud at position B is the most suspect. See Figure 2 for ball stud location and Figure 3 for detail of ball studs.
   - If **ANY** of the ball studs are not mushroomed over, replace the windshield wiper mounting arm and pivot shaft assembly. Refer to the service procedures in section 501-16 of the appropriate Workshop Manual.
   - If all of the ball studs are mushroomed over, continue the inspection procedure.



LOCATION OF BALL STUDS/SOCKETS

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FIGURE 2



RIVETED (MUSHROOMED) **NOT RIVETED**

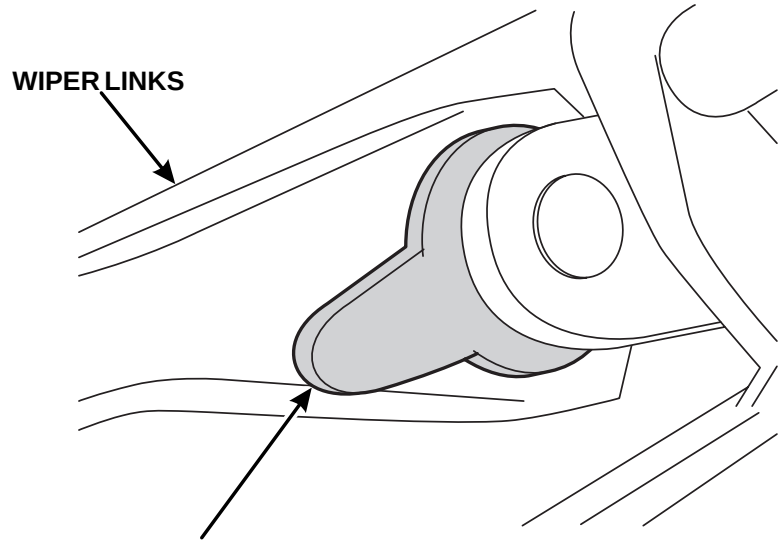
DETAIL OF BALL STUDS

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FIGURE 3

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7. Visually inspect the tabs on the four (4) plastic ball sockets located at the ends of the wiper links. Each tab should be undamaged and fully engaged in the corresponding hole located in the wiper arm link. See Figure 2 for location of ball sockets and Figure 4 for detail of tabs.



EXAMPLE OF GOOD BALL SOCKET
(TAB UNDAMAGED AND FULLY ENGAGED IN WIPER LINK HOLE)

00S49D

FIGURE 4

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A. If the tab is missing, damaged, or detached from the rim on ANY of the sockets, replace the windshield wiper mounting arm and pivot shaft assembly. Refer to the service procedures in section 501-16 of the appropriate Workshop Manual. See Figure 5.

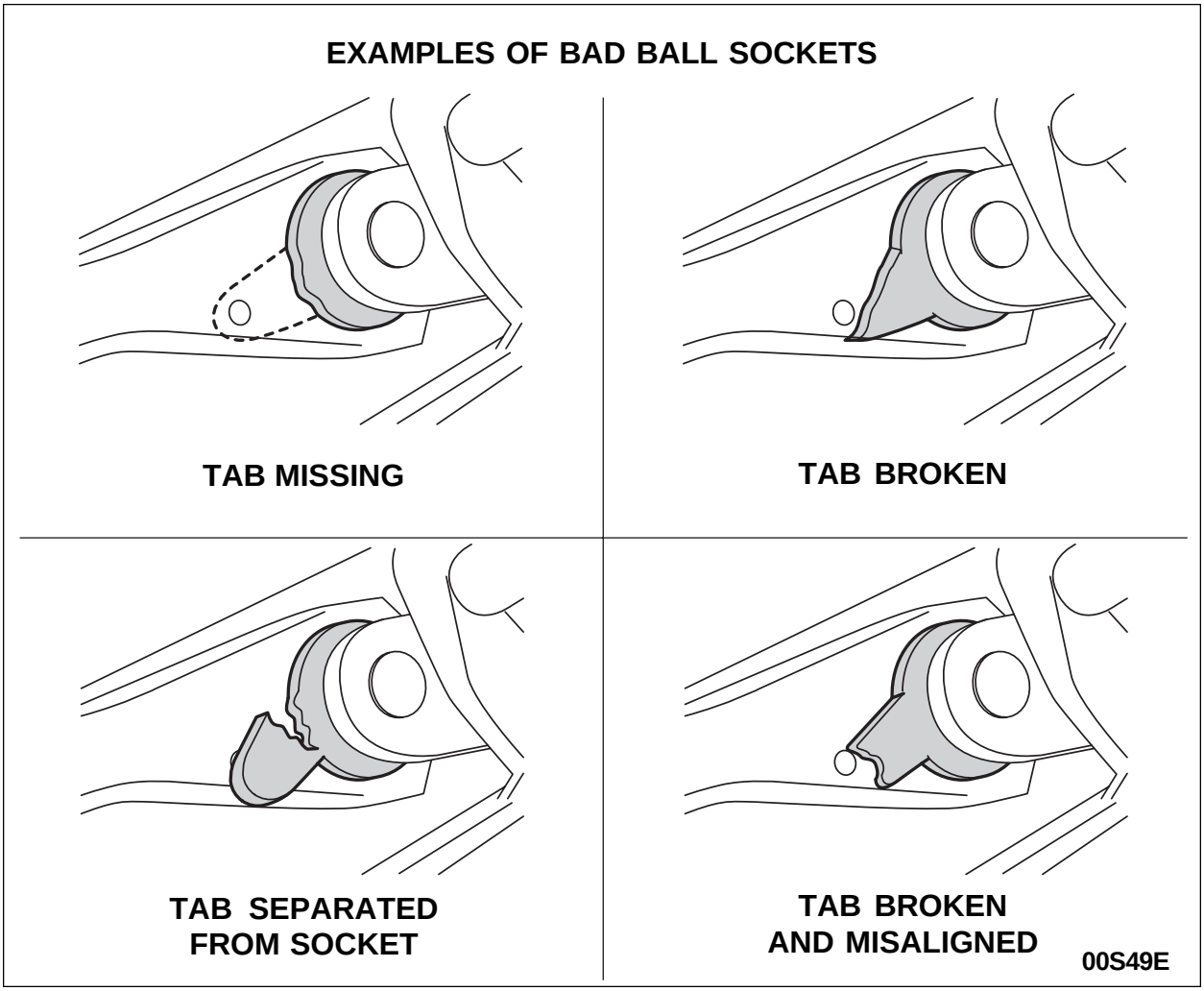
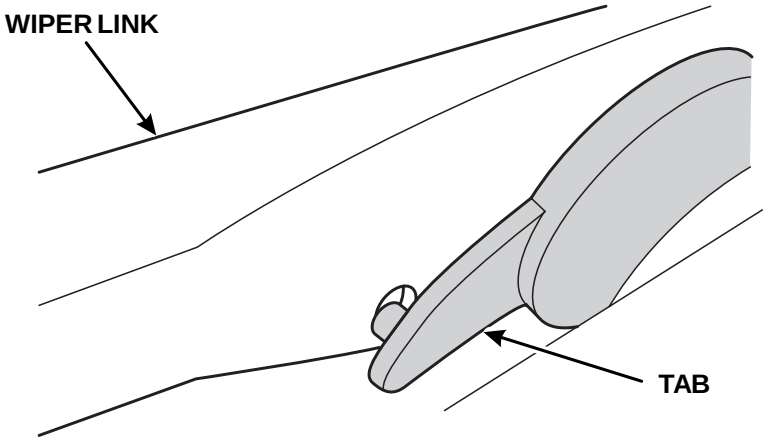


FIGURE 5

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B. If the tab is aligned but not fully engaged into the corresponding hole in the link on **ANY** one of the sockets but is not otherwise damaged or detached from the rim, press the tab with your thumb (no tools) to fully engage it in the hole. If force greater than thumb pressure is required to engage the tab into the hole, replace the windshield wiper mounting arm and pivot shaft assembly. Refer to the service procedures in section 501-16 of the appropriate Workshop Manual. See Figure 6.

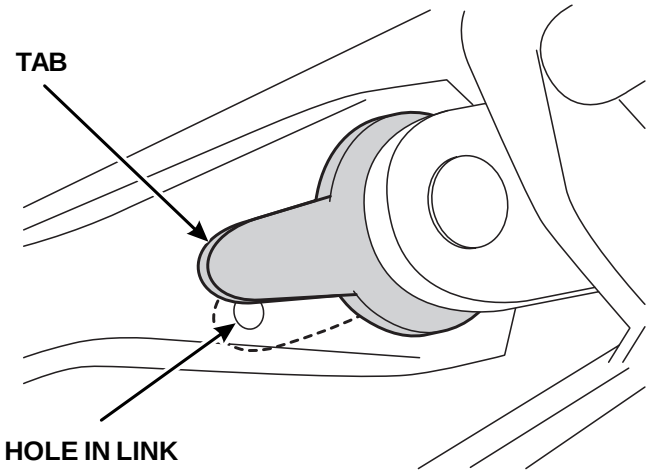


TAB IS ALIGNED WITH HOLE BUT NOT ENGAGED

00S49F

FIGURE 6

C. If the tab is not aligned with the corresponding hole in the link on **ANY** one of the sockets, replace the windshield wiper mounting arm and pivot shaft assembly. Refer to the service procedures in section 501-16 of the appropriate Workshop Manual. See Figure 7.



TAB IS NOT ALIGNED WITH HOLE

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FIGURE 7

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D. Beginning at the tab, carefully lift the lip of the rubber seal using a flat screwdriver or similar tool. Verify that there is a continuous rim of plastic material around the base of **EACH** socket. If there are any sections of material broken away or any visible damage (cracks) on **ANY** one of the sockets, replace the windshield wiper mounting arm and pivot shaft assembly. Refer to the service procedures in section 501-16 of the appropriate Workshop Manual. See Figure 8.

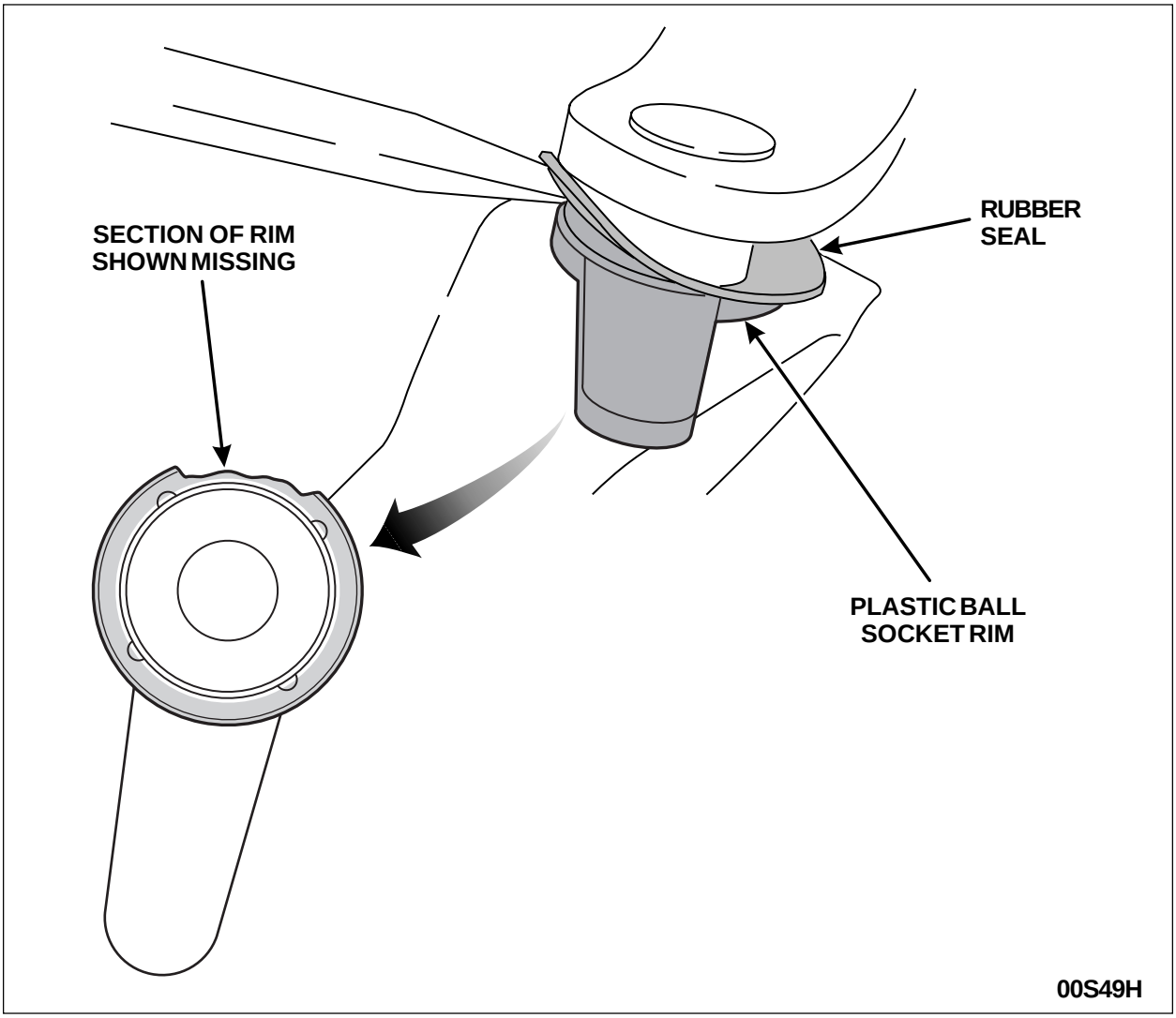


FIGURE 8

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E. Inspect the dome section of the sockets for cracks. If grease is present on the dome of the socket, this may be evidence of a crack. If **ANY** one of the sockets is cracked, replace the windshield wiper mounting arm and pivot shaft assembly. Refer to the service procedures in section 501-16 of the appropriate Workshop Manual. See Figure 9.

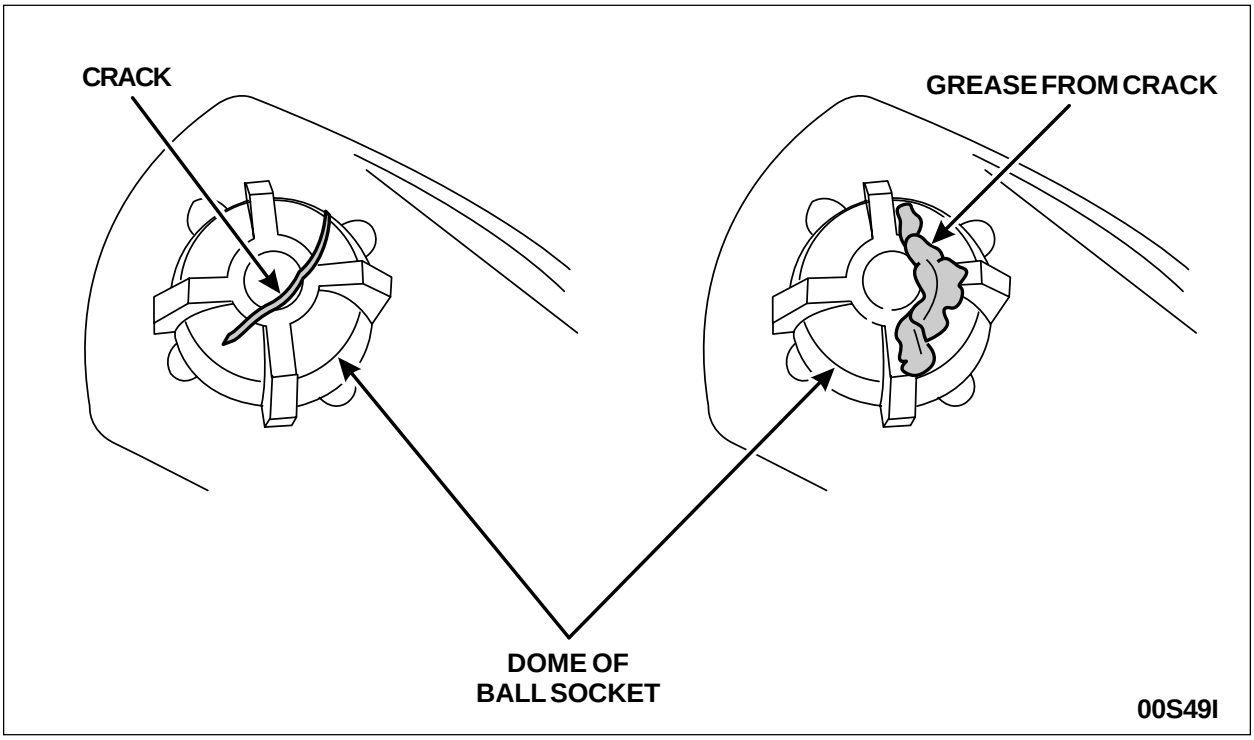


FIGURE 9

If the wiper module passes all the inspections described in A-E, no replacement is required.

8. Install the wiper module. Tighten the bolts to 12 Nm (107 lb-in).
9. **WARNING: THE IGNITION KEY MUST BE IN THE "OFF" POSITION BEFORE CONNECTING THE WIPER MOTOR. IF NOT, SUDDEN MOVEMENT OF THE WIPER LINKS MAY OCCUR WHEN THE MOTOR IS CONNECTED.**

   Connect the wiper motor electrical connector.
10. Install the left side cowl vent panel.
11. **CAUTION: The wiper motor must be in the "PARK" position before installing the wiper arms or damage to the vehicle will result.**

    Cycle and park the wiper motor.
12. **CAUTION: Proper torquing of the wiper arm nuts is critical.**

    Install the windshield wiper arms. Tighten the nuts to 35 Nm (26 lb-ft).

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